When Do Metastable Wavefronts Bind Experience Across Time?

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Abstract

Conscious contents unfold amid neural events separated by spatial propagation, recurrent processing, and measurement delays. Static synchrony, sustained activation, representational similarity, and decoder accuracy can establish that information is present, decodable, or available for behavioral control, but they do not by themselves establish either causal binding across time or phenomenal unity. This distinction is especially relevant after a preregistered adversarial comparison found no sustained posterior synchronization of the form predicted in the tested integrated information theory account and more limited prefrontal ignition and content representation than predicted in the tested global neuronal workspace account [3]. The present paper asks under what conditions a metastable neural wavefront—traveling activity that remains for a finite interval within a specified tolerance of a recoverable propagating family before escaping—can be treated as a candidate temporal-binding mechanism rather than merely as traveling activity correlated with access, report, or task control. The framework uses arrival-aligned moving sections and route-isolated intervention kernels defined relative to a declared neural scale, a specified family of distributed feature contrasts, and explicit interventions that block or condition on designated alternative routes and common drivers. A raw higher-order Möbius contrast is not sufficient to identify conjunction or binding, because such a contrast can remain nonzero for independent feature channels under general response parameterizations. Accordingly, the relevant quantity is an interaction residual evaluated against an explicit no-interaction baseline and interpreted only within the declared intervention model; without the interventions and assumptions needed to estimate that residual, it remains unidentified. The Moving-Frame Binding Conjecture states that a metastable wavefront supports temporally unified conscious content only if its arrival-aligned, route-isolated higher-order intervention kernel remains transport-equivalent across a finite metastable interval, rejects phase-scrambled, standing-wave, feedforward-replay, and common-driver nulls, and changes nongauge operator class at independently anchored content transitions while local feature encoding remains intact. Transport equivalence is assessed only after kernels at successive moving sections have been mapped into compatible intervention and response spaces, avoiding an untyped identification of operators defined on different sections. A fixed-intervention underdetermination construction shows that observed trajectories, sampled representational geometries, fixed-clock decoders, synchrony summaries, access behavior, report distributions, and selected immediate intervention responses can be observationally matched by systems that differ in cross-time causal transport. Thus, cross-time transport is an additional evidential target rather than a consequence of static structural correspondence, decodability, access, or pointwise causal effects. Transported intervention geometries further motivate a gauge-reduced criterion for content transition: a transition is nongauge when no admissible semantics-preserving relabeling and coordinate transformation aligns the pre-

and post-transition operator families. Such a result would identify a change in causal-dynamical organization, not an intrinsic quale. The conjecture therefore supplies a necessary-condition test for a candidate temporal-binding mechanism under explicit modeling assumptions; it is not sufficient for phenomenal consciousness, does not infer phenomenality from access or control, and does not identify what any experience is intrinsically like.

Research Question and Hypothesis

Research question

When does a metastable neural wavefront implement a causally transported temporal-binding mechanism rather than merely accompany correlated traveling activity? More specifically, under what declared neural scale, feature family, intervention semantics, route specification, and readout family can one identify a higher-order intervention kernel that is preserved across moving neural sections, discriminates transport from local recreation and common drive, and changes at independently anchored content transitions?

This mechanistic question does not by itself identify phenomenal unity, phenomenal character, or intrinsic qualia. At most, the proposed criterion identifies a candidate neural mechanism for temporally coordinating distributed content. Whether that organization is constitutive of, necessary for, or merely correlated with unified experience requires an additional phenomenal bridge.

The question therefore separates six targets that are often conflated:

1. **Phenomenal consciousness** concerns whether there is something it is like for a system and, if so, the character and apparent temporal unity of that experience. It is not identified solely by neural transport, behavioral availability, or report.
2. **Access or control** concerns the availability of content for inference, memory, flexible behavior, action selection, or other declared downstream uses. Arrival-aligned causal influence on such readouts can support an access-and-control claim without establishing phenomenality.
3. **Self-report** is one behaviorally and neurally mediated output channel. It may be contemporaneous, delayed, indirect, distorted, or absent. Report can provide evidence under auxiliary assumptions, but it neither defines consciousness nor transparently reveals intrinsic qualitative character.
4. **Perceived consciousness** is an observer's attribution of consciousness to another system. Such attribution may depend on behavior, appearance, theoretical commitments, or social context and is not equivalent to the target system's phenomenal state.
5. **Agency** concerns endogenous action selection, authorship, and control. It is not equivalent to temporal binding, access, reportability, or phenomenal consciousness.
6. **Moral status** is a normative property requiring further premises about sentience, interests, welfare, standing, or rights. No wavefront statistic, intervention kernel, or operator-equivalence result entails it.

Moving-Frame Binding Conjecture

Mechanistic conjecture. Relative to a declared neural scale and family of distributed feature contrasts, a metastable wavefront supports temporally unified conscious content only if its arrival-aligned, route-isolated higher-order intervention kernel remains transport-equivalent across a finite metastable interval, rejects phase-scrambled, standing-wave, feedforward-replay, and common-driver nulls, and changes nongauge operator class at independently anchored content transitions while local feature encoding remains intact.

The principal terms are operationally constrained as follows:

- **Arrival alignment** expresses interventions and responses relative to the independently estimated arrival of the candidate wavefront at each neural section, rather than merely aligning observations by laboratory time or choosing a post hoc time warp that maximizes similarity.
- **Route isolation** requires interventions capable of distinguishing influence transmitted along the candidate route from effects independently recreated at successive locations, propagated through an alternative route, or induced by a common driver. The intervention alphabet and target semantics must remain fixed across compared sections. If the available interventions do not distinguish these possibilities, the route-specific kernel is **unidentified**, even if a traveling pattern can be estimated precisely from passive recordings.
- **Higher-order intervention kernel** denotes the response to declared joint feature interventions after accounting for a preregistered null that permits arbitrary marginal feature effects, independent-channel products, and lower-order compositions. A nonzero Möbius or factorial signed-measure term alone does not identify conjunction, interaction, or binding, because such a term can remain nonzero when feature channels change independently. Evidence for a higher-order interaction therefore requires a departure from the appropriate factorized or lower-order interventional baseline. Even such a departure identifies only an intervention-relative interaction structure, not phenomenal binding by itself.
- **Transport equivalence** requires the section-specific kernels to belong to the same operator class under a declared transport map and a declared family of admissible dynamic gauge transformations. Admissible gauges may relabel latent coordinates or equivalent representations only when they preserve the declared intervention targets, readout predictions, section ordering, and temporal composition. If the gauge family is left unrestricted or selected after observing the result, operator-class preservation is not identified.
- **Metastability** denotes a finite interval over which the relevant operator class is approximately preserved and displays bounded recovery from specified transverse perturbations, followed eventually by escape or transition. The interval, tolerance, perturbation class, and escape criterion must be fixed independently of the equivalence result; persistent oscillation or visual regularity alone is insufficient.
- **Null rejection** requires interventionally discriminating predictions. A phase-scrambled control tests whether the result depends on the relevant phase relations; a standing-wave control tests whether propagation is required; a feedforward-replay control tests whether a matched sequence without the candidate recurrent or transported causal organization suffices; and a common-driver control tests whether successive section responses are independently generated by shared input. Descriptive superiority over these controls is not equivalent to causal rejection when the relevant interventions are absent.
- **Nongauge operator-class change** means that no admissible gauge transformation maps the pretransition kernel to the posttransition kernel. The transition must be anchored independently of the operator analysis, for example by a declared stimulus, task, behavioral, or decoding criterion. Such anchors may establish a transition in content access or control; they do not, without a phenomenal bridge, establish a transition in experience.
- **Preserved local feature encoding** requires the declared local feature information to remain recoverable under the same measurement and validation rules across the transition. This control is intended to distinguish a change in distributed transport organization from a generic loss of signal, arousal, responsiveness, or feature representation.

The conjecture is a proposed **necessary condition for wavefront-mediated temporal binding**, conditional on the declared scale, content family, interventions, routes, readouts, nulls, tolerances, and gauges. It is not a sufficient condition for phenomenal consciousness. A standing process, nonpropagating recurrent assembly, or another dynamical mechanism might perform temporal binding instead. Conversely, a system could satisfy the proposed neural criterion and exhibit robust access and control while lacking phenomenality if the required phenomenal bridge is false. The criterion therefore identifies, at most, a candidate binding mechanism; it does not identify intrinsic qualia or establish that transported neural structure is itself experiential.

Evidence and inference standard

Claims below are typed as follows:

- **Observed evidence:** a reported empirical result or documented data resource, including its measurement and design limitations.
- **Formal consequence:** a result following from explicitly stated mathematical assumptions, intervention semantics, null models, and equivalence relations.
- **Mechanistic inference:** an interpretation that could explain the evidence but requires causal discrimination from specified alternatives.
- **Identified quantity:** a target uniquely determined, up to the declared equivalence relation, by the available observational and interventional distributions under stated assumptions.
- **Unidentified quantity:** a target for which two or more causally or phenomenally distinct models remain compatible with the available evidence. An unidentified quantity is reported as unidentified rather than assigned a preferred value from descriptive fit.
- **Phenomenal bridge:** an additional premise connecting neural organization, access, or control to the presence, unity, or character of experience.
- **Speculation:** a possibility not presently supported by discriminating evidence.

Descriptive fit is not treated as causal explanation. In particular, traveling phase structure does not establish transported causal organization; a nonzero higher-order signed-measure contrast does not establish conjunction or binding; successful decoding does not establish route-specific control; and access, control, or report does not establish phenomenality. Identifiability—the question of whether the target is determined in principle by the available observations and interventions—is separated from estimation, which concerns recovery from finite, noisy data. Where route isolation, intervention semantics, gauge restrictions, or phenomenal bridges are absent, the corresponding mechanistic or phenomenal target remains unidentified.

Specialist Framework and Contribution

Why neural dynamics changes the unit of analysis

A neural-dynamics perspective replaces a static state, region, scalar integration score, or fixed-time graph with a process having propagation lineage, endogenous timing, direction, local stability, recruitment, and escape. The primary object is therefore not "activity in region v at laboratory time t," but a transient path through neural tissue whose different components are reached at different physical times.

Let V be a declared set of neural recording or model sites, and let $X_v(t)$ denote the state measured or modeled at site $v \in V$ and physical time t. A candidate wavefront W has an **arrival field**

$$T_W : U_W \subseteq V \to \mathbb{R},$$

where $T_W(v)$ is the independently estimated time at which the same propagating event reaches site v. For wave phase s, define the moving section

$$F_s = \{(v,t) : v \in U_s,\ t = T_W(v) + s\}.$$

The active domain U_s may change with s. This allows recruitment and release of neural components rather than forcing the mechanism into a fixed subsystem boundary.

A valid arrival field is not any smooth time warp that improves a fit. It must satisfy empirical constraints:

- spatially connected propagation lineage;
- physiologically interpretable direction and delay;
- stability under reasonable changes in filtering and reference;
- out-of-sample generalization across trials or participants;
- and superiority to volume-conduction, common-evoked-response, edge, and arbitrary-registration controls.

A standing oscillation has spatial phase structure but no net propagation lineage through successive sections. Sequential activation may have an arrival order but need not carry a causal state between locations. Global synchrony may have neither. These are distinct dynamical objects even when they produce similar power or phase-locking summaries.

Metastability as finite residence, recovery, and escape

"Metastable" is not used here as a synonym for variability, synchrony, or long duration. Let Z_s be the neural state after registration to the moving section F_s, and let $\mathcal{M}_W$ be a family of coherent propagating states. For a declared tolerance ϵ, define the escape phase

$$\tau_{\text{esc}} = \inf\{s : d(Z_s, \mathcal{M}_W) > \epsilon\},$$

where d is a prespecified state-space distance.

A candidate process is metastable over an interval $I = [s_0, s_1]$ only if three properties hold:

1. **Finite residence:** the process remains near $\mathcal{M}_W$ across a nontrivial interval with probability exceeding a declared threshold.
2. **Transverse recovery:** after a bounded perturbation δZ_s away from the propagating family,

$$\mathbb{E}\big[d(Z^{\delta}_{s+\Delta}, \mathcal{M}_W)\big] \le \rho\, d(Z^{\delta}_s, \mathcal{M}_W) + \eta, \qquad 0 \le \rho < 1,$$

for a declared recovery phase Δ and noise floor η.

1. **Eventual escape:** escape remains possible, and the residence time is not infinite merely because the stimulus or analysis window is sustained.

A fragile feedforward transient can satisfy neither recovery nor residence. A stationary attractor can satisfy recovery but not finite escape. Either process might still contribute to consciousness, but neither is a metastable wavefront under this definition.

The conjunction-sensitive binding operator

Suppose m distributed feature channels have been independently calibrated. Each channel $i \in \{1, \dots, m\}$ has a baseline value 0 and contrast value 1. For each subset $S \subseteq [m]$, let

$$K^{S}_{W;s,h}(dr)$$

be the route-isolated probability law of a downstream neural or control readout R on section F_{s+h}, when the feature channels in S are set to contrast and all remaining channels are set to baseline on F_s. "Route-isolated" means that the causal interpretation is conditional on declared cuts, clamps, conditioning variables, or other evidence capable of excluding relevant bypass and common-driver routes. It does not imply that present passive datasets already provide such isolation.

Define the m-way Möbius interaction

$$B_W(s,h) = \sum_{S \subseteq [m]} (-1)^{m-|S|} K^{S}_{W;s,h}.$$

Because the terms are probability kernels, $B_W(s,h)$ is generally a signed measure. For two features,

$$B_W = K^{\{1,2\}} - K^{\{1\}} - K^{\{2\}} + K^{\varnothing}.$$

This removes the immediate effects attributable to either feature alone. A nonzero norm $\|B_W(s,h)\|$ therefore indicates a conjunction-sensitive response under the declared interventions. It is not automatically perceptual binding: nonlinear saturation, a downstream threshold, or an unblocked common pathway can also generate a nonadditive response.

The criterion consequently requires **local feature preservation**. If a perturbation reduces B_W, each constituent feature must remain detectably encoded at the relevant local sites. Otherwise the apparent loss of binding is equally explained by destruction of the inputs.

Longitudinal composition

Let $P^0_{W;s_1,s_2}(dz_2 \mid z_1)$ be the route-isolated baseline transition kernel carrying a section state from F_{s_1} to F_{s_2} , with no new feature intervention between the two sections. Its action on a signed measure ν is

$$(P^0 \circ \nu)(A) = \int P^0(A \mid z)\, \nu(dz).$$

A transported interaction should satisfy, approximately,

$$B_W(s, h_1 + h_2) \approx P^0_{W;s+h_1,s+h_1+h_2} \circ B_W(s, h_1).$$

The corresponding composition defect is

$$\epsilon_{\text{comp}} = d_{\text{ker}} \left(B_W(s, h_1 + h_2), P^0_{W;s+h_1,s+h_1+h_2} \circ B_W(s, h_1) \right),$$

where d_{ker} is a declared distance between finite signed kernels, such as total variation in a finite-state model. Low defect says that the conjunction-sensitive effect detected downstream is causally descended from the earlier effect. It does not merely recur at the correct latency.

Moving-frame causal covariance

Let $\phi_s : X(F_s) \to Z_s$ map a microstate on section F_s to a proposed macrostate. Let $P^a_{s,h}$ be the micro-level transition under intervention a, and let $\bar{P}^a_{s,h}$ be the proposed macro transition. The arrival-aligned commutation defect is

$$D_{\text{move}}(s,h) = \mathbb{E}_x \sup_a d\Big[(\phi_{s+h})_\# P^a_{s,h}(\cdot \mid x), \bar{P}^a_{s,h}(\cdot \mid \phi_s(x))\Big] ,$$

where $(\phi_{s+h})_\#$ denotes the pushforward of a distribution into macro coordinates.

A low defect at one cut is not sufficient. The macro process must also reproduce compositions across successive sections. Otherwise temporal averaging can make an independently replayed sequence look causally consistent at every snapshot.

Dynamic gauge

Measurements made at different sites, times, participants, or modalities may use different coordinates. An **admissible gauge transformation** is a relabeling or coordinate change that preserves independently calibrated intervention meanings, source roles, readout roles, and arrival-order constraints.

Examples that may be admissible include:

- an invertible change of measurement basis;
- a global shift of wave phase;
- relabeling among intervention channels demonstrated to be functionally exchangeable;
- and section-registration changes within independently bounded timing uncertainty.

The following are not automatically gauge:

- an arbitrary smooth time warp selected to maximize the effect under test;
- independent content relabeling at every section;
- reversal of source and target when direction has been externally anchored;
- or a transformation that changes which neural population was perturbed.

The specialist contribution is thus not another wave-coherence score. It is a framework in which **higher-order causal interaction, propagation lineage, moving-section composition, metastable recovery, finite escape, and residual gauge freedom are jointly necessary for wavefront operator status**.

Evidence and Literature Synthesis

Current multimodal consciousness evidence

Observed evidence. The preregistered Cogitate adversarial collaboration studied 256 participants using multiple measurement modalities. Some findings aligned with predictions from each theory under comparison, whereas the absence of sustained posterior synchronization challenged a prominent IIT prediction and limited prefrontal ignition and content representation challenged prominent GNWT predictions [3]. These results constrain those particular predictions. They do not positively identify traveling-wave binding, establish phenomenal absence when a predicted signal is absent, or show that posterior and prefrontal activity exhaust the relevant dynamical alternatives.

Three open resources support common-data reanalysis. The intracranial EEG resource contains recordings from 38 patients performing the Cogitate visual-perception paradigm [4]. A multi-center MEG-EEG resource provides temporally resolved recordings under the shared protocol [6]. A multi-site fMRI resource contains data from 118 participants, together with behavioral, eye-tracking, quality-control, and analysis records [5]. These publications are peer-reviewed data descriptors. They do not report that any traveling-wave statistic is specific to consciousness, temporally unified content, or phenomenality.

The modalities constrain different quantities. iEEG provides high temporal resolution and direct local field measurements but samples clinically determined sites. MEG-EEG offers broader temporal coverage, while estimates of cortical propagation remain sensitive to source reconstruction, field spread, volume conduction, and reference choices. fMRI constrains coarser spatial organization and network recruitment but cannot directly resolve propagation over millisecond neural delays. Cross-modal convergence should therefore be assessed through modality-appropriate estimates of arrival order, direction, transition timing, spatial recruitment, and finite residence, rather than by treating voltage, magnetic-field, and hemodynamic measurements as interchangeable observations of one wave variable.

The available datasets can support descriptive analyses of phase gradients, arrival fields, cross-time generalization, and their relationships to stimulus conditions, performance, confidence, and report. They do not by themselves provide route-isolated causal kernels. Route isolation here requires a declared intervention regime in which a target perturbation is applied while specified alternative pathways or common drivers are blocked, clamped, randomized, or otherwise separately identified under jointly admissible conditions. If those manipulations are absent, the contribution transported along a proposed route remains unidentified, even when a propagation-like sequence is reliably decoded.

Metastability likewise cannot be inferred from prolonged activity alone. In this framework it denotes persistence, over a prespecified finite interval and at a declared neural scale, of an arrival-aligned transport relation within stated tolerances, together with bounded variation or recovery under the admissible perturbations. The interval, tolerances, spatial scale, feature family, and criteria for entry and escape must be fixed independently of the episode being classified. Without those declarations, “metastable” remains a descriptive label rather than an identified dynamical property.

Nonlinear access is not yet temporal binding

Backward-masking work reported a nonlinear neural threshold associated with changes in objective performance and subjective reports as the target–mask interval increased [21]. Such a threshold is compatible with recurrent amplification or another dynamical transition. Its operational anchors, however, primarily concern access-and-control: availability of stimulus information for discrimination, decision, memory-guided use, action, or report under the task. They do not by themselves determine whether the transition marks phenomenal presence, global availability, decision formation, confidence construction, or report preparation. Phenomenality—the occurrence and character of experience—therefore remains distinct from the access and control capacities measured by these outcomes.

A wavefront account makes a more specific prediction than “late nonlinear activity occurs.” It requires a change in an arrival-aligned, intervention-defined transport operator while the constituent feature representations remain operationally detectable. A threshold in scalar amplitude, decoding accuracy, behavioral performance, or report probability could occur without transport of a stable higher-order causal effect. Conversely, detecting local feature information does not establish that those features are phenomenally present or temporally bound.

A nonzero higher-order inclusion–exclusion or Möbius contrast cannot, by itself, identify conjunction or binding. For example, if two intervention-sensitive output channels remain completely independent, with joint kernels $K_{ab} = P_a \otimes Q_b$, then

$$K_{11} - K_{10} - K_{01} + K_{00} = (P_1 - P_0) \otimes (Q_1 - Q_0),$$

which can be nonzero despite the absence of any interaction between the channels. Such a contrast records a higher-order difference in the chosen signed-measure representation; it is not a binding witness merely because it is nonzero. Its interpretation depends on the output space, intervention semantics, factorization null, and route structure.

The evidential target is therefore not nonzeroness alone. The candidate operator must remain transport-equivalent across a finite metastable interval, survive route-isolated perturbational tests, and reject phase-scrambled propagation, standing waves, feedforward replay, and common-driver alternatives. At an independently anchored content transition, it must change nongauge operator class while local feature encoding remains intact. Here “nongauge” means that the change cannot be removed by an allowed relabeling or coordinate transformation that preserves the declared interventions, readouts, arrival relations, and route metadata. Even satisfaction of these conditions would identify only a candidate mechanism for binding distributed content across time. It would not be sufficient for phenomenal consciousness and would not identify intrinsic qualitative character.

Traveling-wave evidence and its limits

A 2026 bioRxiv modeling preprint by Behler and colleagues reports that hippocampal excitability gradients can generate waves above a critical gradient threshold, that propagation direction can reverse with frequency, and that anterior-to-posterior and posterior-to-anterior waves can have different modeled effects on cortical organization [11]. The study is non-peer-reviewed, model-based, and does not investigate phenomenal consciousness. Its relevance is correspondingly limited: within the model, downstream organization depends not only on local activity magnitude but also on the gradient, direction, and propagation history that generate the activity.

Mechanistic inference. These modeled direction-specific consequences make propagation direction a candidate mechanistic variable, not an established causal variable in the human neural systems considered here. The consciousness bridge remains unsupported unless direction-sensitive dynamics are linked to independently discriminated content transitions under interventions that distinguish transport from common input, generic excitability changes, memory-related dynamics, arousal, volume-conducted fields, and analysis-induced phase gradients. Direction or phase progression alone is insufficient, because feedforward sequences, standing-wave mixtures, phase-scrambled replays, and spatially delayed common drive can produce propagation-like observations without transporting the same causal effect.

Structural correspondence and its ceiling

Representational similarity analysis compares representational dissimilarity matrices while abstracting away from a direct correspondence between recording channels, model units, or measurement modalities [20]. This abstraction is valuable for comparing neural, behavioral, and computational organizations. It does not generally identify causal ancestry or transport. Two systems can exhibit the same representational dissimilarity matrix at each sampled time while differing in whether later states inherit a perturbational effect from earlier states, arise from independent feedforward inputs, or are jointly generated by a hidden driver. Pointwise representational correspondence can therefore preserve relational geometry while erasing temporal lineage.

Neurophenomenal structuralism proposes that qualities may be individuated through their positions in a relational quality space and explores structure-preserving mappings between phenomenal and neural spaces [8]. A methodological defense of structuralism distinguishes empirical structural matching from stronger metaphysical conclusions about what qualities intrinsically are [9]. Formal discussions of mathematical structures of experience likewise clarify the kinds of objects that quality-space approaches seek to compare, although the cited record is a non-peer-reviewed preprint [7].

These approaches support disciplined comparison of relations, but the ceiling on inference must remain explicit. Similarity, isomorphism, or intervention-preserving equivalence between neural structures can identify only the structure encoded by the declared variables and maps. It cannot recover spatial embedding, causal lineage, route identity, or direction if those properties were omitted from the compared objects. Nor does a neural–phenomenal structural match identify which intrinsic qualities, if any, occupy the matched positions unless an additional psychophysical or metaphysical bridge is supplied.

IIT 4.0 advances a stronger identity claim connecting maximal cause–effect structures with phenomenal existence [10]. The present framework does not assume that identity. Even if an arrival-aligned wavefront operator were causally identified up to the declared operational gauge, an inference to phenomenal existence or intrinsic phenomenal character would require an additional bridge principle. Without such a principle, phenomenality and intrinsic qualia remain unidentified.

Critical engagement with the two recent lab papers

Both lab papers considered here are AI-generated speculative manuscripts with AI peer review, not human-peer-reviewed scientific authorities. They are used as formal interlocutors rather than as empirical evidence.

Nassar's causal-island framework distinguishes three questions: whether a sufficiently rich intervention-context profile identifies recurrent causal organization, whether a restricted empirical profile captures the relevant isomorphisms, and whether residual automorphisms leave phenomenal labels unidentified [1]. The distinction among causal organization, operational gauge, and phenomenal underdetermination is retained.

Its limitation for the present conjecture is that a fixed island is not the appropriate object when the candidate mechanism is defined by propagation through changing spatial sections. A finite wavefront model can recruit and release sites, cross an anatomical partition, undergo direction-dependent leakage, collide with another front, or escape from the interval under study. In such a model, boundary crossing may be part of the stipulated mechanism rather than a failure of closure. Moreover, an exact profile of all incoming context morphisms is an oracle-level mathematical object, not an experimentally measured collection of intervention distributions. If spatial embedding, arrival fields, route identity, direction, and escape are omitted from the represented object, an isomorphism of that object cannot identify those omitted quantities.

The present framework therefore **transforms** the fixed causal island into a finite wave tube with arrival-indexed sections and potentially variable membership. It **retains** the separation between causal identification and phenomenal labeling. It **rejects** the assumption that approximate closure around a fixed subsystem is the appropriate invariant for every candidate neural mechanism. None of these modifications converts organization-level identification into evidence of phenomenality.

Baner's macro-causation paper usefully distinguishes predictive lumpability from intervention-consistent causal abstraction: a macro variable can forecast behavior while failing to reproduce the downstream consequences of interventions on its micro realizers [2]. That predictive–causal distinction is retained.

Clock-aligned commutation at a fixed cut is not, however, generally the relevant invariant for delayed propagation. If different sites occupy different propagation phases at one laboratory time, a valid delayed relation can fail a simultaneous-time comparison. Conversely, temporal or spatial averaging can make a phase-scrambled replay, standing-wave mixture, or common-driver sequence appear pointwise consistent. Pointwise commutation therefore does not determine longitudinal transport, transverse recovery, direction-dependent escape, or whether the same perturbational effect persists across successive sections.

The replacement is arrival-aligned causal covariance with explicitly typed composition. If $T_{s\to s'}$ maps intervention effects or state distributions from section s to section s', and $T_{s'\to s''}$ maps from s' to s'', the composite $T_{s'\to s''} \circ T_{s\to s'}$ is defined only when the first map's codomain matches the second map's domain, or when an explicit identification or transport map supplies that match. No equation may silently treat section-specific state spaces, perturbation alphabets, or output measures as the same object. Failure to establish the required typing leaves longitudinal composition unidentified rather than falsified or confirmed.

Intervention semantics must also remain fixed across the identification argument. A global intervention on a feature variable and a section-local perturbation of neural state are not interchangeable merely because both affect the same recorded output. They can be related only by an explicit realization map showing how the feature intervention is implemented at the relevant section while preserving feature meaning and admissibility. If that map is unavailable, evidence from one intervention family cannot identify the kernel associated with the other. The resulting framework therefore restricts comparisons to physically interpretable neural manipulations with declared targets, timing, routes, and feature semantics; positive propagation delay is treated as an endogenous causal coordinate rather than as a clock-alignment nuisance.

This paper thus **transforms** fixed-cut commutation into arrival-aligned causal covariance plus typed longitudinal composition. It does not infer binding from a nonzero Möbius contrast, and it does not infer phenomenality from successful causal abstraction. Category-theoretic machinery is not required for the operative test because the central empirical issue is whether cross-time causal transport has been measured under a coherent intervention family rather than omitted by the chosen representation.

Evidence hierarchy

Evidential tier	What it presently supports	What it does not support
Preregistered multimodal results	Constraints on particular preregistered IIT and GNWT predictions under the tested paradigms	A positive wavefront mechanism, exhaustive theory comparison, or a direct measure of phenomenality
Open iEEG and MEG-EEG	Descriptive tests of arrival fields, phase gradients, timing, finite residence, and cross-time generalization	Route-isolated causal transport, exclusion of all propagation nulls, or consciousness-specific binding
Open fMRI	Coarse spatial organization, recruitment, and network-level constraints	Direct identification of millisecond-scale neural propagation or arrival-aligned intervention kernels
Masking and report studies	Candidate nonlinear transitions in performance, report, and other access-and-control measures	A clean separation of phenomenal presence from access, decision, confidence, and report preparation

Evidential tier	What it presently supports	What it does not support
Neural-field modeling preprints	Model-level plausibility of gradient-, frequency-, and direction-dependent waves	Empirical validation in the relevant neural systems, route-isolated transport, or consciousness-specific conclusions
Higher-order intervention contrasts	Signed-measure differences across combinations of interventions when intervention semantics are fixed	Conjunction or binding from nonzeroness alone; independent channels can also yield nonzero contrasts
Structuralist methods	Disciplined comparison of representational or relational geometries	Causal lineage unless encoded, or identity of neural structure with intrinsic phenomenal character
AI-generated lab theories	Formal distinctions, counterexamples, and candidate assumptions for further analysis	Established scientific evidence, empirical validation, or authority for claims about phenomenality

Cross-Disciplinary Integration

From static quality spaces to transported discrimination geometry

For each content condition c, let $B_W^c(s,h)$ denote the higher-order signed intervention kernel at section s and arrival-aligned horizon h. The term "higher-order" here describes the algebraic order of the intervention contrast; it does not establish conjunctive encoding, statistical dependence, or binding. In particular, a Möbius contrast can be nonzero for independent feature channels. In the two-feature case, if the joint response law factorizes under every intervention as

$$\mu_{ij} = \alpha_i \otimes \beta_j,$$

then

$$\mu_{11} - \mu_{10} - \mu_{01} + \mu_{00} = (\alpha_1 - \alpha_0) \otimes (\beta_1 - \beta_0),$$

which is generally nonzero despite complete factorization. Thus B_W^c measures a joint higher-order change in the response law, not dependence or conjunction by itself. A binding interpretation additionally requires a statistic or model comparison whose value is known to vanish under the declared independent-channel null, or independent evidence that the transported effect changes a conjunctive readout while the component feature encodings remain intact. Rejection of phase-scrambled, standing-wave, feedforward-replay, and common-driver nulls does not by itself reject independent co-transport of feature effects.

Subject to that restriction, an operator-contrast geometry can be defined by

$$D_{W;s,h}(c,c') = d_{\mathrm{ker}}\left(B_W^c(s,h), B_W^{c'}(s,h)\right).$$

This expression is meaningful only when the content conditions share a declared intervention domain and compatible outcome spaces, and when d_{ker} is defined for the resulting signed kernels. If those conditions are not met, $D_{W;s,h}$ is not identified. The content labels must also be anchored independently of the operator comparison—for example by a prespecified stimulus contrast, task variable, or validated behavioral transition—rather than inferred from the same kernel geometry that they are then used to interpret. Such anchors concern experimentally discriminable content or report; they do not by themselves establish phenomenality.

This geometry differs from a conventional neural representational dissimilarity matrix in two respects. First, its points are intervention-defined response operators rather than observed activation patterns. Second, comparison across sections requires an explicitly typed transport relation. Let

$$B^c_W(s,h) : \mathcal{I}_s \longrightarrow \mathcal{M}_\pm(Y_s)$$

map a section-specific intervention domain $\mathcal{I}_s$ to signed measures on an outcome space Y_s. For two arrival-aligned sections, let

$$\mathsf{A}_{s\to s'} : \mathcal{I}_s \longrightarrow \mathcal{I}_{s'}$$

identify corresponding interventions, and let

$$\mathsf{T}_{s\to s'} : \mathcal{M}_\pm(Y_s) \longrightarrow \mathcal{M}_\pm(Y_{s'})$$

transport their outcome measures. Longitudinal equivalence is then a comparison between maps with the same domain and codomain:

$$B^c_W(s',h') \circ \mathsf{A}_{s\to s'} \;\simeq_G\; \mathsf{T}_{s\to s'} \circ B^c_W(s,h).$$

Here $\simeq_G$ denotes agreement within a prespecified tolerance after quotienting only by a declared gauge G. The intervention correspondence, transport map, gauge, and tolerance must be fixed or estimated independently of the particular content transition being tested. If they are selected post hoc to maximize agreement, transport equivalence becomes non-diagnostic. A gauge transformation is restricted to an invertible reparameterization that preserves intervention identities, arrival order, and all tested outcome probabilities. It cannot be used to erase a difference between null and target conditions or to identify operators that make different interventional predictions.

Route isolation is likewise an interventional requirement rather than a property inferred from propagation latency alone. At the declared neural scale, the estimated kernel must concern the proposed route while relevant alternative paths and common drivers are experimentally blocked, randomized, conditioned on under justified assumptions, or otherwise distinguished by the design. When those conditions cannot be met, the route-specific transport kernel—and therefore the corresponding moving-frame comparison—remains partially or wholly unidentified. Observing an orderly wave, lagged predictability, or section-to-section decoding is insufficient to identify it.

Formal consequence. Identical snapshot geometries do not determine transport. Distances within each section specify neither $\mathsf{A}_{s\to s'}$ nor $\mathsf{T}_{s\to s'}$, and distinct longitudinal couplings can preserve the same collection of within-section distances. Conversely, transport-equivalent higher-order contrasts need not encode dependence: two independent feature channels can be carried together by the same propagation map.

Phenomenal bridge. If phenomenal similarities and transitions depend partly on the structure of temporally transported neural discriminations, then an independently identified operator geometry may constrain hypotheses about phenomenal organization. This is a bridge hypothesis, not a consequence of the neural formalism.

Limit of the bridge. Structural matching does not select intrinsic qualitative labels, identify a subject, or establish that the neural process is experienced. An interventionally effective wave may organize access, memory, decision, or report without being sufficient for phenomenality. The methodological restraint defended in structuralist work therefore remains necessary [9].

Temporal unity and the specious present

A finite metastable residence interval is a candidate mechanism for maintaining neural causal continuity across physically separated events. Here metastability should be defined relative to a declared scale, state-space neighborhood, and tolerance: over a finite interval, the process remains within the specified dynamical regime and its arrival-aligned transport discrepancy remains below the prespecified threshold before escape or transition. The interval and threshold are not determined by the label “metastable” and must be estimated independently of the content-transition test.

Even when identified, this residence interval should not be equated with the duration of the phenomenological “specious present.” Residence time, oscillatory period, conduction delay, behavioral integration window, memory availability, report latency, and experienced duration are distinct variables. A correspondence among them would require independent psychophysical and neural evidence. Without such evidence, the duration and boundaries of the phenomenal interval remain unidentified.

The framework also does not require temporal unity to mean strict physical simultaneity. Its mechanistic proposal is narrower: events reached at different times may participate in one extended neural process when their feature-sensitive consequences remain in the same route-isolated, recoverable propagation lineage. Establishing that lineage would support causal continuity and possibly temporally extended access or control. It would not, without an additional bridge, establish that the events compose one phenomenal episode.

An independently anchored content transition is particularly important here. A nongauge change in operator class has evidential force only when the transition time is determined without using that same operator change, and when local feature encoding remains measurable across the transition. Otherwise, an apparent change of class could reflect relabeling, loss of the component features, a change in measurement sensitivity, or an analysis boundary rather than a change in temporal binding.

Predictive processing and precision routing

Predictive-processing approaches seek systematic mappings between neural processes and phenomenological structure [25]. Within such approaches, a measured traveling pattern could reflect prediction-error propagation, changing gain, precision weighting, or routing among hierarchical levels. Those possibilities do not establish that the wavefront binds content, but they provide alternative mechanistic interpretations that can be compared interventionally.

The accounts can be related only under an explicit mapping assumption. If precision dynamics causally alter the route-isolated transition kernels that transport content-sensitive effects, then a wavefront may implement one component of a predictive process. If precision changes coordinate distributed readouts through another pathway while the measured wave is epiphenomenal or redundant, the wavefront fails the present causal criterion. If both descriptions refer to different scales of the same interventionally identified process, they may be complementary rather than competing.

The relevant comparison is therefore not whether predictive-processing terminology can redescribe a wave. It is whether perturbing the proposed precision mechanism changes the arrival-aligned transport kernel, whether perturbing the wave route changes the predicted downstream effects, and whether the resulting model rejects the same phase-scrambled, standing-wave, feedforward-replay, and common-driver nulls. Even a successful comparison would identify causal organization relevant to access and control. It would not establish phenomenality or intrinsic qualitative character.

Dreaming and sensory-independent contents

Research on the neural correlates of dreaming is relevant because post-awakening reports provide evidence about content under conditions in which contemporaneous exteroceptive input is reduced or differently related to reported experience [23]. Such findings do not show that dream contents are carried by the perceptually anchored wavefront proposed here, nor do they provide the route-isolated intervention kernels required by the conjecture.

A sensory-anchored arrival field therefore cannot be assumed to generalize to dreaming. Failure to observe the same perceptual wavefront in dream-related measurements would bound the proposed mechanism to exteroceptive perceptual binding rather than refute every dynamic account of temporal unity. Conversely, evidence for an internally generated recurrent wave tube would not be sufficient merely because its trajectory resembles a waking wave. It would require a declared internal arrival coordinate, route-specific perturbation, rejection of replay and common-driver alternatives, preservation of local content encoding, and independently anchored timing of content transitions.

That final requirement is difficult when content is reconstructed retrospectively from awakening reports. In such cases, the timing of within-dream content changes, the relevant metastable interval, and the separation of dream generation from memory and report preparation may remain unidentified. Neural classification or later report can establish discriminability and access under a measurement model, but neither alone determines whether the classified state was phenomenal or when its phenomenal boundaries occurred.

Neural binding versus intentional binding

The intentional-binding literature investigates how intentions, expectations, and causal beliefs relate to reported temporal attraction between actions and outcomes [14] [15]. That operational phenomenon concerns judgments of action–outcome timing and agency. The present operator concerns the causal transport of distributed neural feature effects. The shared word “binding” does not imply a shared mechanism.

A change in reported temporal attraction is therefore not evidence for moving-frame neural binding unless a specific causal link is established. The change could arise from perceptual timing, expectation, decision, memory, response calibration, or some combination of these factors. Conversely, a route-isolated transport effect need not alter intentional-binding reports if it concerns features or pathways not used for action–outcome judgments.

A productive connection would require an explicit model in which interventions on the proposed wave route alter both the relevant transport kernel and the temporal judgment in a predicted manner, while interventions on alternative routes dissociate them. Even then, the result would primarily identify causal access to, or control of, temporal judgments. It would not by itself show that phenomenal time had contracted, that two events were phenomenally unified, or that the neural operator constituted their qualitative temporal relation.

Approximate closure and multiscale causal atlases

Centered Closure Theory emphasizes approximately closed grains, interventional epistemology, and automorphism-limited quality comparison, but it is an AI-generated, non-peer-reviewed working draft and introduces centering and intrinsic geometry without independent empirical validation [48]. Multiscale

Reflexive Causal Geometry likewise separates mechanism, bridge, and metaphysical layers and proposes intervention-defined causal atlases, but its gluing, self-indexing, and phenomenal geometry remain theoretical components of an AI-generated working synthesis rather than established results [49]. These drafts can supply comparative vocabulary, not empirical support for the present conjecture.

For present purposes, approximate closure is scale- and intervention-relative. A neural grain is approximately closed only to the extent that, over a declared interval and intervention family, unmodeled external perturbations do not change its estimated internal transition kernel beyond a prespecified tolerance after measured boundary inputs are taken into account. Neither the grain nor its degree of closure is identified merely by observing strong within-grain connectivity. Boundary perturbations and comparisons across plausible grains are required, and more than one grain may remain compatible with the available data.

The useful idea retained from both drafts is multiscale comparison under constrained gauge. A wave tube may traverse spatial, temporal, and descriptive scales, but the claim that its local sections "glue" must be expressed through typed intervention correspondences and transport maps whose diagrams commute within a declared tolerance. Pairwise resemblance among local state spaces is insufficient. If several incompatible atlases fit the same observational and interventional data, the multiscale organization remains non-unique.

Gauge freedom is also limited by empirical invariance. Transformations may identify coordinate descriptions that preserve all declared intervention and outcome relations, but they may not identify kernels with different route-specific predictions or quotient away an independently anchored content transition. A change in nongauge operator class is therefore a change that survives all admissible reparameterizations, not simply a large distance in an arbitrarily chosen representation.

No centering principle follows from transport equivalence, approximate closure, or successful multiscale gluing. Those properties could identify a causally organized system capable of maintaining and routing information for access, control, memory, or report. They do not determine that the system is a subject, that its states are phenomenal, or that an operator geometry is an intrinsic quality geometry. The Moving-Frame Binding Conjecture accordingly identifies, at most, a candidate causal mechanism for temporally extended neural binding under its declared scale, intervention family, route-isolation assumptions, and null tests; phenomenality and intrinsic qualitative identity remain outside what the transport criterion identifies.

Analysis, Argument, or Model

Qualification ledger for a wavefront binding operator

A candidate wavefront W earns binding-operator status only relative to a declared neural scale, family of distributed feature contrasts, intervention alphabet, arrival field, and route-isolation regime. Let

$$K^{W}_{j\to k} : \mathcal{Y}_j \rightsquigarrow \mathcal{Y}_k$$

denote the arrival-aligned controlled kernel from moving section F_j to moving section F_k. "Route-isolated" means that, under the declared intervention regime, directed paths from the intervention target to the downstream readout that do not belong to the candidate wavefront route are cut, clamped, or otherwise excluded, and the intervention does not alter common drivers. If those conditions cannot be established, the total intervention effect may still be estimable, but the wavefront-specific kernel is unidentified.

A further qualification concerns the signed-measure Möbius contrast. For binary feature interventions, a contrast of the form

$$B_W(s,h) = \sum_{a \in \{0,1\}^m} (-1)^{m-|a|} K^W_{s \to s+h}(\cdot \mid do(A=a))$$

is a legitimate mixed intervention-response contrast, but $B_W(s,h) \neq 0$ does not by itself identify conjunction, statistical dependence, or binding. In particular, if the feature channels are completely independent and

$$K^W_{s \to s+h}(\cdot \mid do(A=a)) = \bigotimes_{i=1}^{m} K_i^{a_i},$$

then

$$B_W(s,h) = \bigotimes_{i=1}^{m} \left(K_i^1 - K_i^0\right),$$

which is generally nonzero whenever each independent channel responds to its own intervention. The contrast therefore records a joint mixed effect across intervention conditions, not an interaction beyond independent channel effects.

Higher-order nonseparability must instead be defined relative to a prespecified null family. Let $\mathfrak{N}_{<m}$ be the declared family of kernels generated by independent feature channels, lower-order interactions, and any admissible nuisance transformations. Define

$$J_W(s,h) = \inf_{L \in \mathfrak{N}_{<m}} D\left(K^W_{s \to s+h}, L\right),$$

where D is a prespecified distance or divergence evaluated on held-out intervention conditions. A positive J_W rejects only the declared no-higher-order model; it does not establish a universal or semantics-independent notion of conjunction. If $\mathfrak{N}_{<m}$, the feature decomposition, or the intervention semantics are not fixed independently of the fitted wavefront model, higher-order nonseparability is unidentified.

Subject to these qualifications, the candidate must satisfy all six conditions below.

Condition	Required evidence	Failure interpretation
W1. Propagation lineage	An independently estimated arrival field, directed connected path, and delays within declared physiological bounds, with registration assessed out of sample	Apparent gradient, standing wave, arbitrary time warp, or lineage unresolved
W2. Higher-order nonseparability	$J_W(s,h)$ exceeds estimation uncertainty and matched lower-order or separable nulls; a nonzero B_W alone is insufficient	Independent channel effects, lower-order structure, misspecified null family, or higher-order structure unresolved
W3. Feature preservation	Constituent features remain locally decodable or interventionally discriminable when J_W or the transported operator class changes	Input destruction, generalized state loss, or measurement failure rather than selective loss of candidate binding

Condition	Required evidence	Failure interpretation
W4. Longitudinal composition	Direct route-isolated kernels agree, within a prespecified tolerance, with compositions of successive moving-section kernels over a finite interval	Common-driver replay, parallel-route contamination, omitted state, nonstationarity, or failure of the controlled-Markov assumptions
W5. Metastability	A prespecified finite residence-time regime, recovery from bounded transverse perturbations before exit, and a nonzero escape process on the observation scale	Fragile feedforward transient, stationary attractor, or metastability unidentified
W6. Null rejection	The moving-frame result distinguishes the candidate from matched standing-wave, feedforward-replay, phase-scrambled, common-driver, and registration nulls while preserving the nuisance statistics assigned to each null	The evidence does not discriminate transport from one or more adequate alternatives

Longitudinal composition must be type-consistent. For $i < j < k$,

$$\left(K^W_{j\to k} \circ K^W_{i\to j}\right)(E \mid y_i) = \int_{\mathcal{Y}_j} K^W_{j\to k}(E \mid y_j)\, K^W_{i\to j}(dy_j \mid y_i),$$

so both the direct kernel $K^W_{i\to k}$ and the composed kernel map $\mathcal{Y}_i$ into $\mathcal{Y}_k$. A corresponding composition defect may be defined by

$$\epsilon_{\mathrm{comp}}(i,j,k) = D\left[K^W_{i\to k}, K^W_{j\to k} \circ K^W_{i\to j}\right].$$

This comparison is justified only when the section states are causally sufficient at the declared scale, intervention meanings are stable across sections, and the controlled transition kernels are modular and Markov at that scale. A low defect is evidence for transport equivalence under those assumptions, not an assumption-free proof of a physical carrier.

Metastability is likewise scale-relative. Let $\mathcal{M}_W$ be a prespecified neighborhood of the candidate traveling regime in the chosen coarse state space. Evidence for metastability requires a residence time longer than local relaxation, recovery toward $\mathcal{M}_W$ following bounded perturbations transverse to the propagating direction, and exits from $\mathcal{M}_W$ on a finite longer timescale. A long-lived trajectory without perturbational recovery does not distinguish metastability from a slow transient, while recovery without an identifiable escape process does not distinguish it from a stationary attractor.

The six conditions jointly identify a candidate causal-dynamical mechanism for temporal binding. They are not a consciousness score, do not establish phenomenality, and do not identify intrinsic qualitative character.

Snapshot–Transport Underdetermination

Assumptions

Consider a finite sequence of moving sections $F_0, \ldots, F_n$, all represented here by the same finite state space $\mathcal{Y}$ for simplicity. Let:

- $A = (A_1, \ldots, A_m) \in \{0,1\}^m$ be distributed feature inputs;
- H be a latent trial variable containing stimulus context and shared noise;

- $Y_k \in \mathcal{Y}$ be the neural state recorded at section F_k;
- $q(A, H)$ be an arbitrary finite-valued neural response map;
- O be a terminal access, control, or report output;
- and $I_k^{\lambda_a}$ be a section-local intervention that replaces the mechanism for Y_k with a draw from a prespecified distribution λ_a, indexed if desired by feature pattern a.

The intervention has the same semantics at every section and in both systems: it replaces only the local Y_k mechanism and leaves A, H, and all other structural equations unchanged. A deterministic state write is the special case $\lambda_a = \delta_{y^*(a)}$. The construction does not require q to contain a genuine higher-order interaction. If a higher-order immediate profile is required, the intervention family $\{\lambda_a\}$ may be chosen to have $J > 0$ relative to the declared null family.

Coherent transport system C

The first section is initialized by

$$Y_0 = q(A, H),$$

and successive sections inherit the preceding state:

$$Y_{k+1} = Y_k, \qquad k = 0, \ldots, n-1.$$

Under $I_k^{\lambda_a}$, the structural equation for Y_k is replaced by

$$Y_k \sim \lambda_a.$$

All later sections then inherit that intervened state through the chain

$$Y_k \to Y_{k+1} \to \cdots \to Y_n.$$

Common-driver replay system R

The common exogenous variables directly set every section:

$$Y_k = q(A, H), \qquad k = 0, \ldots, n,$$

with no causal arrows from Y_k to Y_{k+1}. The resulting sequence is a coordinated common-driver replay rather than a transported lineage. Under $I_k^{\lambda_a}$, only the equation for Y_k is replaced. Later sections remain directly determined by A and H.

Observational equivalence

Under passive observation, the two systems can be coupled so that

$$(Y_0, \ldots, Y_n)_C = (q(A, H), \ldots, q(A, H)) = (Y_0, \ldots, Y_n)_R$$

almost surely. They consequently have the same complete observed trajectory law, not merely the same one-time marginals. They also have the same:

- recorded state at every sampled section;

- representational dissimilarity matrices and other sampled state-space geometries;
- fixed-clock and section-specific decoder performance;
- power, synchrony, covariance, and time-averaged summaries computed from the recorded trajectory;
- observational transition conditionals;
- and passive distributions of any terminal O defined by the same function of A, H, and the observed trajectory.

Equality of such access, control, or report distributions establishes only equivalence with respect to the modeled outputs. Phenomenal state is not a variable in the construction and is therefore not identified by this equivalence.

Pointwise intervention equivalence

For any section F_k, apply the same local replacement intervention $I_k^{\lambda_a}$ and read out Y_k before any later section evolves. Both systems then implement the same immediate law:

$$\mathcal{L}_C(Y_k \mid I_k^{\lambda_a}) = \lambda_a = \mathcal{L}_R(Y_k \mid I_k^{\lambda_a}).$$

Thus the complete family of immediate section-local intervention kernels may be identical. If $\{\lambda_a\}$ has a nonzero Möbius contrast or a positive nonseparability residual J, both systems exhibit that same immediate profile at the intervened section. Immediate higher-order structure therefore does not determine whether that structure is transported.

Longitudinal intervention difference

Now apply $I_k^{\lambda_a}$ while route isolation ensures that the intervention does not alter A, H, or any alternative route to a later section F_ℓ, where $\ell > k$.

In the coherent system,

$$\mathcal{L}_C(Y_\ell \mid I_k^{\lambda_a}) = \lambda_a, \qquad \ell > k,$$

because the intervened state is inherited by every downstream section. For a deterministic write,

$$do(Y_k = y^*) \quad \Rightarrow \quad Y_\ell = y^* \qquad \text{for all } \ell > k.$$

In the replay system,

$$\mathcal{L}_R(Y_\ell \mid I_k^{\lambda_a}) = \mathcal{L}(q(A, H)), \qquad \ell > k,$$

because later sections are reset by the common driver and do not descend causally from Y_k. Consequently, the later response in R is independent of the intervention index a, whereas the later response in C retains any dependence carried by λ_a. If $\{\lambda_a\}$ has a positive higher-order residual relative to $\mathfrak{N}_{<m}$, that residual is transported in C and absent downstream in R. A nonzero Möbius contrast may likewise be transported in C, but its interpretation remains a mixed intervention effect rather than, by itself, evidence of conjunction.

Formal result

Snapshot–Transport Underdetermination result. Under the finite-state construction above and the declared local replacement intervention semantics, complete observational trajectories, sampled state-space geometries, local decoders, synchrony summaries, passive access or report distributions, observational transition conditionals, and immediate section-local intervention kernels do not determine whether an intervention-indexed neural effect is longitudinally transported. The coherent transport and common-driver replay systems agree on all those quantities by construction but differ in their local-to-later intervention kernels.

The result applies equally when the immediate intervention family has a prespecified higher-order nonseparability residual: local nonseparability and longitudinal transport are distinct properties. Evidence that resolves the difference must constrain the longitudinal causal path, for example through a route-isolated local-to-later intervention kernel or another independently justified path-resolving design.

This is a constructive identifiability result, not an empirical finding and not a claim that every biological replay model can match every wavefront model. It establishes that the listed snapshot and pointwise quantities are insufficient over a model class containing these two systems.

Exact boundary of the result

The underdetermination disappears when the evidence already includes:

1. the route-isolated, intervention-indexed transition kernel between each pair of successive sections;
2. a causally sufficient state representation at the declared scale;
3. stable and modular intervention semantics across sections;
4. and the controlled-Markov and stationarity assumptions required for composition.

Under those conditions, longer-horizon kernels are fixed by the typed Chapman–Kolmogorov relation

$$K^W_{i\to k}(dy_k \mid y_i) = \int_{\mathcal{Y}_j} K^W_{j\to k}(dy_k \mid y_j)\, K^W_{i\to j}(dy_j \mid y_i), \qquad i < j < k.$$

The result therefore does not claim that temporal composition is unknowable. It identifies what passive geometry and immediate cut-local interventions omit. If route isolation, causal sufficiency, or intervention stability is unavailable, the wavefront-specific longitudinal kernel remains unidentified even when an observational transition model can be estimated.

Phase-scrambled and feedforward variants

A within-condition trial permutation can preserve each time point's empirical response distribution, condition means, local decoding, and report frequencies while destroying trial-wise cross-time lineage. Multivariate phase-randomized surrogates can be constructed to preserve selected spectra, and conduction-delay-matched feedforward models can be constructed to preserve selected arrival-time histograms. Unlike the common-driver construction above, these nulls need not preserve the full observational joint distribution. Their role is narrower: each controls a different nuisance structure that might otherwise produce an apparent moving-frame advantage.

No single scramble is sufficient. If a null preserves spectra but changes instantaneous covariance, its rejection may be attributable to covariance. If it preserves covariance but destroys local feature encoding, rejection may reflect content loss rather than loss of transport. If a feedforward null does not match path length, conduction delay, and local response reliability, its failure does not isolate recurrence or metastability. A useful hierarchy must therefore state which statistics each null preserves, verify those preservation claims, and avoid treating rejection of one null as rejection of all nontransport alternatives.

Null rejection also does not repair absent route isolation. A phase-scrambled or feedforward model can be disfavored observationally while the causal contribution of the candidate wavefront remains unidentified because an intervention can still reach the downstream readout through a parallel route or common driver.

Moving versus fixed-clock macro consistency

Suppose sites u and v are reached at times $T_W(u)$ and $T_W(v)$. A fixed-clock cut at time t may compare u after the candidate wave has passed with v before arrival. A propagating macrostate can therefore show a high fixed-clock commutation or composition defect despite a low arrival-aligned defect.

The converse can also occur. A common stimulus can directly evoke site-specific delayed responses whose marginal laws align after temporal averaging even though a perturbation at u never influences v. Low fixed-clock defect is consequently neither necessary nor sufficient for wavefront transport.

Mechanistic implication. If the candidate satisfies the moving-frame criterion, held-out route-isolated intervention kernels should exhibit lower composition error under an independently constrained arrival field than under a fixed-clock registration. An observational moving-frame advantage can support propagation-lineage estimation under W1, but it cannot by itself establish the route-isolated causal transport required by W4. Improvement under an unrestricted smooth time warp is not probative because the registration can absorb arbitrary sequential structure. The arrival field, allowable smoothness, delay bounds, and model-selection procedure must therefore be fixed independently of the held-out comparison.

Gauge-reduced neural qualitative transition

Let $\mathcal{O}^-$ and $\mathcal{O}^+$ be pre- and post-transition neural operator families. Each includes the content-indexed route-isolated kernels, higher-order residual structure relative to $\mathfrak{N}_{<m}$, transition kernels, typed composition relations, and arrival-field metadata over a local interval. Let $\mathcal{G}_{\mathrm{sem}}$ be a prespecified group of invertible, semantics-preserving gauges. Membership in $\mathcal{G}_{\mathrm{sem}}$ must preserve the declared intervention alphabet, feature anchors, temporal orientation, relevant units or metrics, and readout meanings; it cannot be inferred solely by selecting transformations that minimize the transition residue.

Define

$$\mathcal{R}_{\mathrm{tr}} = \inf_{g \in \mathcal{G}_{\mathrm{sem}}} d_{\mathrm{op}} \left(\mathcal{O}^+, g \cdot \mathcal{O}^- \right),$$

where d_{op} is a prespecified composite distance comparing like-typed kernels, higher-order residual maps, arrival structures, and composition relations.

A transition is **neural-nongauge** when the estimated $\mathcal{R}_{\mathrm{tr}}$ exceeds estimation uncertainty and matched nontransition nulls. If the admissible gauges, kernel types, content anchors, or operator distance have not been independently specified, the residue—and hence nongauge status—is unidentified.

In a finite representation, candidate invariant changes include:

- rank changes in the higher-order residual map defined relative to $\mathfrak{N}_{<m}$;
- emergence or loss of a causally transported subspace;
- changes in direction-dependent composition structure;
- changes in prespecified branching, collision, or escape topology;
- and singular-subspace changes, provided the admissible gauges preserve the metric required to interpret singular values.

An amplitude change need not be nongauge if it is absorbed by an admissible rescaling. A content-label permutation is inadmissible when the labels are independently anchored rather than fitted from the same neural data. Conversely, a sensor-basis rotation should not count as a qualitative neural change if it preserves intervention identities, readout semantics, and all operator relations relevant to the criterion.

Formal interpretation. At the population level, $\mathcal{R}_{\mathrm{tr}} > 0$ identifies a change in the declared causal-dynamical operator class: no admissible semantics-preserving gauge maps the pre-transition family to the post-transition family. This conclusion is conditional on the declared operators, metric, anchors, intervention regime, and gauge group.

Phenomenal bridge. A neural-nongauge transition is not thereby a phenomenal transition. If its external anchor is a stimulus discrimination, action policy, access state, or verbal report, then the immediate conclusion concerns that anchored neural, access, control, or report organization. The residue can bear on phenomenal organization only given an additional, independently defended bridge principle linking transported neural discrimination structure to phenomenal structure. Covariation between $\mathcal{R}_{\mathrm{tr}}$ and report or access is not itself such a principle, and the moving-frame criterion remains neither sufficient for phenomenal consciousness nor an identification of intrinsic qualia.

Ceiling. Even perfect identification of $\mathcal{O}$, including its route-isolated transport kernels and nongauge transitions, does not select intrinsic phenomenal labels when the remaining causal automorphisms have no empirically privileged fixed point. Structural correspondence can constrain relations among discriminable contents without determining what, if anything, those contents are intrinsically like. This preserves the separation between empirical structural matching and stronger metaphysical identification emphasized in structuralist methodology, as well as the corresponding limitation in the AI-generated causal-island analysis [9] [1].

Alternatives, Counterarguments, and Boundary Cases

The higher-order intervention kernel must not be interpreted as a stand-alone measure of conjunction, binding, or consciousness. In particular, a signed-measure Möbius contrast can be nonzero even when two feature channels respond independently. If the outcome measures factorize as $P_{ij} = P_i^A \otimes P_j^B$, then

$$P_{11} - P_{10} - P_{01} + P_{00} = \left(P_1^A - P_0^A\right) \otimes \left(P_1^B - P_0^B\right),$$

which is generally nonzero. “Higher-order” therefore describes the inclusion–exclusion order of the intervention contrast, not an identified interaction or conjunction. A binding interpretation requires an explicitly specified independence or additive null, arrival-aligned transport across the declared sections, route-isolated causal effects, rejection of the competing dynamical models considered below, preserved local feature encoding, and a nongauge change at an independently anchored content transition. Even this conjunction of results would identify only a candidate neural binding mechanism, not phenomenal consciousness or intrinsic qualia.

All comparisons must also retain the same intervention semantics. A global intervention on a feature source, a local intervention on the state at an intermediate section, and an intervention on an edge or propagation route define different causal quantities. They cannot be substituted for one another in a longitudinal argument without an explicit causal model linking them. Where that link, the relevant positivity conditions, or the required route exclusions cannot be established, the route-specific quantity is unidentified.

Common drivers

A cortical, subcortical, or exogenous driver could impose orderly latencies across sites while separately influencing downstream access, control, and report. Such a model can be observationally compatible with an apparently coherent traveling trajectory. Arrival order, phase gradients, and successful local decoding therefore do not by themselves distinguish propagation from coordinated responses to a common cause. Nor does transport of a nonzero Möbius contrast resolve the ambiguity, because independent feature responses can already generate such a contrast.

A wavefront receives route-specific causal support only if interventions with fixed semantics alter later, prespecified feature-contrast effects through the nominated propagation lineage. Operationally, route isolation must be defined relative to an explicit causal graph and route: the intervention should alter the relevant edge or lineage while off-route pathways, source-feature semantics, and local feature encoding are preserved within declared tolerances. If an intervention can influence the downstream outcome through the driver, collateral tissue effects, or another recurrent pathway, the path-specific effect is not isolated.

Conditioning on measured driver activity can reduce a modeled confound but is not generally equivalent to intervening on the driver. Residual measurement error, latent driver components, treatment-induced confounding, or unmeasured parallel paths can leave the traveling-mediator and common-driver accounts observationally indistinguishable. In those cases, causal attribution to the wavefront remains unidentified rather than merely uncertain in degree.

Heterogeneous feedforward delays

A heterogeneous feedforward sweep can reproduce arrival order, apparent propagation velocity, local content decoding, and a nonzero higher-order intervention contrast. It can also produce nonlinear downstream responses without implementing recurrent temporal binding. The relevant comparison is therefore not between “a wave” and “no wave,” but between models matched for delays, local encoding, intervention marginals, and downstream readouts.

Metastability must be operationalized at the declared neural scale. The candidate trajectory should remain for a finite, prespecified interval within a declared neighborhood of a moving family of states; bounded perturbations transverse to that family should return toward it before escape; and the residence, recovery, and escape criteria should be specified independently of the result to be classified. A long transient or delay line is not excluded merely by calling it feedforward, and a recurrent process is not established merely by observing persistence.

Longitudinal composition also requires type-compatible objects. Kernels defined over different intervention variables, state spaces, or outcome measures cannot be directly composed without explicit maps between their domains and codomains. In particular, a source-level feature intervention cannot be replaced at an intermediate section by a local state-setting intervention and then treated as the same transported cause. If compatible spaces and pushforward maps have not been specified, the claimed compositional equivalence is unidentified.

A delay-matched feedforward model is excluded only if it fails these independently specified metastability, recovery, type-compatible composition, and route-isolated perturbation tests. Conversely, if a feedforward-replay model satisfies the same arrival-aligned transport equivalence, null rejections, transition-linked operator-class changes, and preservation of local encoding, the proposed experiments have not operationally distinguished it from the wavefront account. The mechanistic classification would then require revision rather than protection by recurrent or wave terminology.

Standing synchrony or nonpropagating recurrence

A standing synchronized assembly or nonpropagating recurrent process could mediate the tested feature coordination while apparent motion arises from cortical geometry, sequential recruitment, spatial filtering, or the measurement reference. Posterior recurrence, workspace-like broadcasting, or another stationary macroprocess could consequently remain necessary after the apparent propagation component is altered.

This alternative makes the gauge specification consequential. The admissible gauge family must state which changes of phase origin, coordinate frame, sensor reference, source basis, or state parametrization count as alternative descriptions of the same physical process. An apparent transition is nongauge only if no admissible transformation maps the pretransition operator into the posttransition operator. The transition must also be anchored by information not derived from the operator-class change itself. An externally controlled stimulus transition or an independently validated content marker may provide such an anchor, but neither by itself identifies a phenomenal transition.

The framework permits a stationary-process result; it does not assume that all candidate binding mechanisms must travel. If selective alteration of propagation direction, continuity, or lineage leaves the prespecified higher-order and transition-linked effects equivalent within an established sensitivity bound, while intervention on a standing process abolishes those effects with local feature encoding intact, the evidence favors the standing process for the tested causal function. It would not, without an additional bridge, establish that the standing process constitutes phenomenal binding.

Workspace and posterior-integration interpretations

A wave could implement communication needed for flexible downstream availability, recurrent posterior differentiation, both, or neither. Mapping it onto a workspace requires evidence that the transported feature information becomes available across the relevant downstream systems and can guide multiple forms of control. Those are operational properties of access and use; they do not by themselves establish phenomenality.

Mapping the process onto posterior integration requires more than posterior localization or synchronization. The pertinent causal organization would have to depend on the wave process rather than on a stationary posterior assembly, a common driver, or local recurrent computations that merely coexist with propagation. A nonzero higher-order kernel is insufficient for this purpose unless the appropriate factorized and stationary-process nulls are rejected.

The mixed Cogitate results constrain particular preregistered predictions of global neuronal workspace and integrated-information accounts, but they do not license replacing either account with a wave mechanism by default [3]. Nor do those results identify the route-isolated intervention kernel proposed here. The present framework instead states conditions that a propagation-based implementation claim would need to satisfy while leaving the relation between access, causal integration, and phenomenality open.

Precision cascades

A precision, gain, or neuromodulatory cascade could coordinate feature processing while a traveling phase pattern reflects only the cascade's timing. The alternatives are separable only if propagation lineage and precision provenance can be manipulated or otherwise identified independently. One test would alter the nominated route while preserving the modeled precision signal and local feature encoding; the converse test would alter precision while preserving the wave's route, arrival structure, and feature interventions.

If those manipulations have collateral effects or if precision and propagation are generated by the same unmeasured process, the decomposition is not identified. In that case, "wavefront binding" and "hierarchical precision control" remain competing descriptions rather than established causal roles. A correlation between phase propagation, gain, and behavioral access cannot determine which process mediates the tested effect.

Seizure-like synchrony, anesthesia, and other altered states

Boundary-test proposal. Anesthesia, seizure-like activity, and disorders of consciousness are not phenomenally homogeneous categories, and behavioral unresponsiveness cannot be treated as a direct measurement of absent experience. These labels also do not by themselves establish that the required metastable wavefront or intervention kernel is present. Comparisons must therefore characterize the neural dynamics, arousal, local feature encoding, sensory responsivity, access capacity, medication or pathology, and measurement sensitivity separately.

The full criterion—not wave presence, synchrony, or persistence alone—can be compared across awake perception and independently characterized altered-state conditions. Relevant questions include whether the same arrival-aligned and route-isolated kernel is transport-equivalent over the declared interval, whether it rejects the same phase-scrambled, standing-wave, feedforward-replay, and common-driver nulls, and whether its nongauge class changes at independently anchored feature or content transitions while local encoding remains intact.

Finding the same feature-contrast-indexed operator in an unresponsive condition would defeat any claim that the operator is sufficient for wakefulness, report, or behavioral access. It would not by itself show that the condition lacks phenomenality, and therefore would not alone refute the operator's more limited status as a candidate binding mechanism. Conversely, failure to detect the operator in one altered state could reflect a genuine mechanistic difference, altered excitability, disrupted local coding, changed neuromodulation, poor intervention efficacy, or reduced measurement sensitivity.

Accordingly, neither presence nor absence in these states identifies phenomenal consciousness. Such comparisons are useful for dissociating the candidate mechanism from arousal, responsiveness, and report only when those variables and the relevant causal estimands are modeled independently.

Dreaming and imagery

Dreaming and imagery need not be organized by an externally driven sensory arrival field. The conjecture could nevertheless be tested by declaring an endogenous event anchor and asking whether internally generated feature contrasts enter a transport-equivalent moving frame with the same route-isolated and transition-linked properties. The endogenous intervention or anchor must not be retrospectively defined from the trajectory whose existence is being tested.

If the operator is not detectable during report-verified dreaming or imagery despite adequate signal sensitivity, preserved local feature encoding, and successful detection of the relevant endogenous events, the evidence would favor restricting the proposal to perceptual temporal binding. Retrospective or prospective reports provide evidence about access to dream or imagined content, but they do not directly identify the full phenomenal state at every tested time point. Conversely, detecting recurrent activity in these conditions would not justify relabeling it as a wave tube unless it passes the same preregistered transport, route, null, metastability, and nongauge-transition criteria.

Minimal adaptive and homeostatic systems

A minimal adaptive network could, in principle, exhibit propagating control signals, finite residence, perturbational recovery, and escape. It might also generate a nonzero signed-measure Möbius contrast through independent control channels. Such a system may lack a report channel, flexible downstream access, or a self-model by construction, but its phenomenality is not thereby established as absent; rather, phenomenality remains unidentified.

These systems are therefore important null and scope cases. If they satisfy the dynamical criterion, that result shows that the criterion is not sufficient for phenomenal consciousness. If they fail only an access or report test, that failure does not establish absence of experience. Causal coordination, adaptive regulation, access, agency, phenomenal consciousness, intrinsic qualia, and moral standing cannot be inferred as automatic rungs of a single scalar hierarchy.

The access-only objection

The operator might mediate availability for decision, working control, memory, or report rather than phenomenality. This account gains support if operator onset reliably follows independently estimated access commitment, decision formation, or motor preparation, or if it depends on task and reporting demands after local feature encoding and sensory drive have been matched. Such results would identify an association with access-related processing, not prove the absence of earlier experience.

No-report and delayed-report conditions can help separate the operator from immediate motor output, but neither condition directly measures phenomenality. Disappearance of the operator in a no-report condition may indicate dependence on task set, access, memory, or report preparation; persistence may indicate report independence. Neither result alone determines whether the represented content was phenomenally experienced.

If the operator robustly predicts only flexible use and coordinated control, the proposal survives in a bounded mechanistic form: it would describe a candidate causal mechanism for temporally coordinated access. Calling that result phenomenal binding would be unwarranted. Conversely, demonstrating report-independent transport and route-specific effects would remove one access confound but still would not make the kernel sufficient for phenomenal consciousness or identify intrinsic qualia.

The epiphenomenal-wave objection

Traveling activity is common in spatially extended excitable systems, so a wave may reflect tissue geometry or population dynamics without mediating the tested feature coordination. The relevant intervention must alter propagation direction, route, continuity, or arrival relation while preserving the source-feature interventions, measured local feature encoding, and plausible off-route inputs within declared tolerances. Preserved decodability alone is not enough if the manipulation changes unmeasured aspects of the local code.

If the prespecified downstream intervention effects and independently anchored transition profile remain equivalent under such propagation alterations, within an established sensitivity and equivalence bound, the result argues against a necessary causal role for the tested wave lineage in that setting. It does not exclude every propagation mechanism at other scales or through unmanipulated routes. If, by contrast, the downstream effects change selectively and mediation through the nominated route is identified against common-driver and collateral-path alternatives, the epiphenomenal account is weakened.

A correlation between wave strength and accurate report cannot adjudicate this objection, because both may reflect sensory drive, arousal, precision, access, or a common cause. The required evidence is selective causal mediation of the declared feature-contrast effect. Even successful mediation would establish access-and-control relevance more directly than phenomenality and would remain evidence for a candidate binding mechanism rather than an identification of conscious experience itself.

Empirical Implications and Falsification Criteria

What existing passive data can establish

The open Cogitate datasets permit descriptive, predictive, and model-comparative analyses. They do not, without additional interventions and causal assumptions, identify a route-isolated neural intervention kernel.

For iEEG and MEG-EEG, passive analyses can assess:

1. whether candidate arrival fields reproduce across held-out trials, participants, sites, and reasonable preprocessing choices;
2. whether physiologically constrained arrival alignment improves held-out prediction or representational stability relative to fixed-clock alignment and flexibility-matched arbitrary-warp controls;
3. whether estimated propagation direction predicts later neural organization after local amplitude, stimulus-locked responses, field spread, and global evoked components are modeled;
4. whether dynamic discrimination geometry is more stable in the declared moving frame;
5. whether spontaneous deviations from the candidate trajectory are followed by observational relaxation toward that trajectory;
6. whether inferred changes in operator class align with independently specified content transitions rather than decision, report, or motor timing; and
7. whether the same descriptive relationships are reproduced at the temporal and spatial resolution appropriate to each recording modality.

These analyses can support the existence of a reproducible traveling pattern and can disfavor specified observational alternatives. They cannot establish that the pattern causes temporal binding, that an apparent recovery is interventionally robust, or that a local-to-later effect travels specifically through the proposed route.

The iEEG dataset is particularly relevant to localized timing but has sparse, clinically determined coverage [4]. MEG-EEG provides broader temporal sampling, but source localization, field spread, reference dependence, and leakage must be treated as model-dependent uncertainties rather than assumed away [6]. Agreement between iEEG and MEG-EEG would strengthen a descriptive propagation claim, but disagreement would need to be interpreted in light of their different sampling and inverse problems.

The fMRI dataset can constrain coarse spatial organization, regional recruitment, and content-related multivariate structure [5]. It cannot by itself identify millisecond arrival fields or determine the direction of fast neural transport. Cross-modal analyses should therefore compare coarse regional ordering, condition-dependent organization, or transition structure under an explicit linking model. Raw hemodynamic and electrophysiological amplitudes should not be treated as measurements of the same instantaneous process.

A further limitation concerns the proposed higher-order contrast. For binary feature interventions $a \in \{0,1\}^m$, the raw signed-measure Möbius contrast has the form

$$B_W^{\mathrm{raw}} = \sum_{a \in \{0,1\}^m} (-1)^{m-|a|} P(Y_W \in \cdot \mid do(F = a))\,.$$

A nonzero value does not identify conjunction, synergy, dependence, or binding. If the output consists of independent channels with

$$P_a = \bigotimes_{j=1}^{m} P_{j,a_j}\,,$$

then

$$B_W^{\mathrm{raw}} = \bigotimes_{j=1}^{m} (P_{j,1} - P_{j,0})\,,$$

which is generically nonzero whenever each independent channel responds to its own intervention. Accordingly, B_W^{raw} is only a mixed intervention contrast: it shows that the joint outcome law changes across the factorial intervention conditions. It is not, by itself, a binding operator.

An empirical binding candidate must instead reject a declared separable causal model of the feature channels and their transport. The resulting nonseparable residual is conditional on the chosen intervention variables, outcome space, route model, and separability null; it is not a model-free quantity. Even rejection of that null would establish only nonseparable causal organization within the tested model class, not phenomenal unity.

If the task does not independently and adequately vary the m required feature contrasts, even the corresponding stimulus-level factorial contrast is unidentified. If the features are randomized factorially, stimulus-to-neural response contrasts may be estimable, but they are not equivalent to interventions on neural sections or routes. Without section-local perturbations or defensible instrumental assumptions, the route-isolated local-to-later kernel remains unidentified. Observational interactions, decoding interactions, and representational-geometry measures may be reported as proxies, but they must not be labeled intervention kernels or causal binding measures.

Required temporal null systems

A defensible analysis should use multiple null systems because no single surrogate preserves all relevant marginals, local codes, timing distributions, and nuisance structure. Each null tests a different component of the proposed account.

Null system	Quantities to preserve or match	Result required to discriminate the candidate account
Within-condition trial permutation	Time-specific marginals, condition means, local decoding, report frequencies, and trial counts	Trial-specific arrival-aligned or longitudinal structure is reduced while matched local content information remains

Null system	Quantities to preserve or match	Result required to discriminate the candidate account
Multivariate phase randomization	Declared power spectra, selected autocorrelation functions, channel covariance targets, and analysis bandwidth	Propagation-lineage or recovery statistics are reduced beyond changes attributable to mismatched local encoding or signal-to-noise ratio
Standing-wave surrogate	Frequency, power, spatial envelope, and, where possible, local phase locking	Net directed section-to-section transport and direction-dependent later effects are absent despite preservation of oscillatory structure
Delay-matched feedforward replay	Arrival histogram, local content encoding, evoked timing, and gross propagation direction	Recurrence-dependent reuse, bounded recovery, collision response, or sensitivity to earlier section perturbation is absent
Common-driver model	The observed trajectory and as much cross-sectional dependence as the fitted model can reproduce	A source- or route-specific intervention changes later sections in a way not reproduced by intervention on the modeled common driver alone
Arbitrary smooth time warp	Comparable alignment flexibility and held-out fitting opportunity	Physiologically anchored arrival fields provide reproducible held-out benefit beyond the penalized arbitrary warp
Evoked-response leakage model	Stimulus-locked latency, common timing, waveform shape, and field-spread structure	Apparent propagation remains after estimated leakage and shared evoked components are removed or explicitly modeled
Local-code-degradation control	Perturbation energy, local signal quality, arousal indicators, and constituent feature decodability	The candidate higher-order residual changes after constituent encoding and measurement sensitivity are equated
Separable-channel intervention model	Marginal intervention responses of each feature channel and their independently transported products	The observed route-specific joint intervention law departs from the declared separable model on held-out interventions or conditions

Failure of a null model to fit does not by itself establish the proposed mechanism. The failure may reflect misspecified noise, unmatched signal-to-noise ratio, destruction of the content signal, excessive surrogate flexibility, or an omitted alternative mechanism. Conversely, a null should not be declared equivalent merely because it matches one summary statistic. Comparisons must use preregistered targets, matched estimation opportunity, held-out evaluation, and explicit tolerances.

The common-driver null has a special status. A common-driver model may reproduce the complete observed trajectory, so passive goodness of fit cannot generally distinguish it from route-specific propagation. Rejection requires an intervention, a valid instrument, or another independently justified source of causal asymmetry. Where none is available, route isolation is unidentified rather than merely noisy.

What stronger causal evidence would require

The framework does not entail a recommendation for any particular experiment, but its causal interpretation requires evidence beyond passive reanalysis. The relevant perturbation would need to alter propagation lineage, direction, continuity, collision behavior, or pinning while approximately preserving local feature encoding, measurement quality, arousal, and nonspecific perturbation energy.

“Route-isolated” must be defined relative to an explicit causal graph or structural model. At minimum, the target kernel must distinguish effects transmitted through the candidate wavefront from effects transmitted through plausible bypass pathways or common drivers. Merely stimulating an early site and observing a later change does not provide route isolation: the effect may travel through another pathway, arise from a common response to the perturbation, or reflect broad state disruption.

Evidence for a candidate transport mechanism would be strongest if:

- the nonseparable intervention residual, rather than the raw Möbius contrast, followed the independently estimated arrival field;
- perturbing an earlier section changed later outcomes at arrival-aligned times, whereas matched perturbations outside the proposed route did not;
- blocking, redirecting, or otherwise changing the candidate route altered the later effect after plausible bypass routes and common drivers were separately modeled or manipulated;
- reversing or disrupting propagation direction changed the later operator structure without comparably degrading constituent feature codes;
- bounded perturbations were followed by recovery within the declared metastable interval, whereas larger or sustained perturbations produced escape under preregistered state and time thresholds;
- the effect generalized to held-out intervention sites, timings, or feature combinations;
- and changes in the candidate operator aligned with independently anchored content transitions rather than being fully explained by arousal, task engagement, decision, access, or report production.

Preservation of local feature decoding is a necessary control within the conjecture, not proof that the underlying local representation is unchanged. Decoder performance can remain stable despite changes in code geometry, noise correlations, downstream readability, or unmeasured features. Local-code preservation should therefore be evaluated using multiple declared measures, including within-site discrimination, cross-condition generalization, signal quality, and, where possible, causal effects on feature-specific behavior.

Likewise, longitudinal composition should be tested only with type-compatible causal maps. If K_{ab} maps interventions or states at section a to probability laws at section b, and K_{bc} maps the resulting admissible laws at b to laws at c, then a comparison with K_{ac} is meaningful only after their domains, codomains, conditioning variables, and transport maps have been made compatible. Pointwise interaction scores or signed measures at separate sections cannot simply be multiplied or composed. Where such typed maps cannot be identified, the longitudinal composition quantity is unidentified; an arrival-ordered correlation is not a substitute.

Direct perturbation of human neural tissue is not implied as a recommendation. Any invasive work would require independent clinical justification, informed consent, risk minimization, and a clear separation of research aims from clinical care. Noninvasive perturbations and naturally occurring clinical interventions may provide partial evidence, but their spatial specificity, state effects, and exclusion assumptions must be stated explicitly.

Formal falsification criterion

The Snapshot–Transport Underdetermination claim must be formulated under one fixed set of variables, observable quantities, intervention targets, and intervention semantics. A valid underdetermination witness requires two models that:

1. agree on all declared static and pointwise observables;

2. agree on the responses to every intervention included among the matched observations;
3. use the same intervention algebra and the same meanings for the intervened variables; and
4. disagree on the target cross-time, local-to-later intervention kernel.

Global interventions on feature variables and section-local interventions on neural states are not interchangeable. A construction that matches models under global feature interventions but distinguishes them using newly introduced section-local interventions does not prove underdetermination for a single declared information set. It instead compares different intervention regimes.

Accordingly, the broad formal underdetermination result remains unestablished unless a same-semantics pair of models or a general nonidentification proof is supplied. This does not make the route-isolated kernel identifiable from the passive datasets. It means that empirical nonidentification from those datasets is a narrower conclusion than a theorem asserting nonidentification under a richer intervention regime.

The formal claim would be falsified by a proof that, under the stated assumptions and fixed intervention semantics, complete matched cut-wise information uniquely determines every target local-to-later kernel. It would also be narrowed if uniqueness followed only after adding assumptions such as causal sufficiency, a known state-space factorization, Markovianity, faithfulness, or exclusion of bypass routes. In that case, identification would be conditional on those assumptions rather than a consequence of the cut-wise observations alone.

Dynamical and mechanistic falsification criteria

For a declared neural scale, feature family, intervention family, gauge group, metastable interval, and error tolerance, the Moving-Frame Binding Conjecture is not supported if any required condition fails under adequately sensitive, held-out or preregistered analysis. Such a failure concerns the declared candidate and scope; it does not show that no traveling neural process could contribute to temporal integration at another scale.

1. **No moving-frame advantage.** Physiologically constrained arrival alignment provides no reproducible held-out improvement over fixed-clock analysis or flexibility-matched arbitrary-warp controls.

1. **No transport equivalence.** After arrival alignment, the estimated route-specific kernels vary across sections or time beyond the declared equivalence tolerance during the proposed metastable interval.

1. **Artifact dependence.** The effect disappears under plausible filtering, reference, volume-conduction, field-spread, evoked-response, edge, site, or source-model controls, without evidence that those controls removed the target signal itself.

1. **Null-system equivalence.** Phase-scrambled, standing-wave, delay-matched feedforward, common-driver, or separable-channel models satisfy the same preregistered transport, recovery, intervention, and transition criteria within uncertainty.

1. **Raw-contrast sufficiency.** The evidence for binding consists only of a nonzero B_W^{raw}, with no rejection of the declared separable intervention model. Because independent channels can produce a nonzero raw contrast, this result does not satisfy the higher-order binding requirement.

1. **Loss of local features.** The putative nonseparable residual disappears after constituent feature decodability, code stability, perturbation energy, signal quality, and relevant state variables are

equated.

1. **Failure of route isolation.** Effects of an earlier perturbation are equally explained by a common driver, broad state change, direct bypass, or nonspecific perturbation response, and the proposed route makes no distinct interventional prediction.

1. **No typed longitudinal effect.** Earlier section perturbations do not change later route-specific kernels at the predicted arrival-aligned times, or the claimed composition relies on operations between incompatible mathematical objects.

1. **No metastability.** Under preregistered neighborhood, recovery, escape, and time criteria, the process exhibits neither bounded recovery after small perturbations nor finite escape under sufficiently large or sustained perturbations. A single transient sweep or a stationary attractor without the declared recovery-and-escape profile does not meet the criterion.

1. **Direction irrelevance.** Direction or continuity can be altered without changing later operator structure, despite adequate manipulation strength, preserved local feature codes, and sufficient measurement sensitivity.

1. **Transition nonspecificity.** Changes in operator class do not align with independently anchored content transitions, or align more closely with stimulus onset, arousal, decision, access, report, or motor events.

1. **Gauge indeterminacy.** The declared gauge group is not specified independently of the result, or the alleged transition residue can be removed by an admissible relabeling that preserves all relevant observational and interventional structure. In that case, a nongauge transition has not been identified.

1. **Alternative mediator superiority.** A standing, nonpropagating, or otherwise distinct process explains the nonseparable effect and mediates its later consequences after the candidate wavefront is altered.

1. **Scale fragility.** Operator status appears only under one unmotivated parcellation, frequency band, reference, source model, temporal window, or time warp and does not survive the declared scale-sensitivity analysis.

1. **Failure of replication.** Arrival fields, transport-equivalence results, or route-specific residuals fail to generalize across held-out trials, participants, sites, intervention timings, or modalities at the spatial and temporal resolution appropriate to each modality.

The relevant thresholds cannot be selected after inspecting the desired result. “Metastable,” “transport-equivalent,” “route-isolated,” and “nongauge” require declared operational definitions. Where those definitions or the necessary interventions are absent, the corresponding quantity is unidentified or untested, not positive by default.

Access and report failure conditions

Content encoding, access-and-control, report production, and phenomenality are distinct explanatory targets. A process may make information available for working memory, decision, flexible control, or action without thereby constituting phenomenal experience. It may also contribute to report production without being the mechanism that made the content accessible.

The proposed mechanism should be classified primarily as report-related if:

- its onset reliably follows decision commitment, response selection, or motor preparation;
- its transition residue tracks report production across content conditions;
- changing report timing or modality changes the operator in parallel with the response plan;
- or its apparent content association disappears after report and motor variables are modeled.

It should be classified primarily as an access-and-control mechanism if:

- it predicts flexible use of content in memory, decision, planning, or action across changes in overt report;
- access accuracy or control demands mediate its association with content judgments;
- it tracks task relevance or global availability more strongly than independently varied content;
- or it is preserved when report production is removed but remains coupled to selection, maintenance, or behavioral use.

Delayed-report and minimized-report conditions can help separate immediate motor production from earlier content-related processing, but they do not isolate phenomenality. Covert decision, memory encoding, expectation, attention, and later report preparation can remain. “No report” therefore means that a particular overt-report confound has been reduced; it does not mean that access, control, or phenomenality has been directly measured.

These outcomes would not make the neural dynamics unimportant. They would change the supported explanatory target from temporal phenomenal binding to report, access, control, or some combination of those functions.

Phenomenal-bridge failure conditions

The Moving-Frame Binding Conjecture identifies, at most, a candidate neural binding mechanism. Satisfaction of its dynamical and causal criteria is not sufficient for phenomenal consciousness and does not identify intrinsic qualitative character. No positive pattern in the present datasets, including a route-isolated nonseparable kernel in a no-report condition, removes that limitation.

A consciousness-specific interpretation would be weakened if:

- the full operator profile were equally present in carefully matched conditions that, under an independently defended operational criterion, differ in conscious-content status;
- the operator were reliably absent in episodes with convergent evidence for conscious content and adequate sensitivity to detect the relevant neural process;
- independently anchored content changes occurred while the gauge-reduced operator class remained stable;
- large nongauge operator changes occurred without corresponding changes in content-related evidence after task, access, report, arousal, and measurement artifacts were controlled;
- or the operator predicted only access, flexible control, decision, or report variables.

These are asymmetric boundary tests rather than direct measurements of phenomenality. Labels such as “unconscious,” “unresponsive,” “no-report,” or “dreaming” do not by themselves establish the presence or absence of phenomenal content. Each comparison requires an independently defended criterion, and residual uncertainty about phenomenality must remain explicit.

A positive association with conscious-content judgments can support a neural correlate or an access-related mechanism. A route-isolated intervention can support causal involvement in the measured neural or behavioral outcome. Neither result alone entails that the mechanism is constitutive of phenomenality. Establishing such a constitutive relation would require an additional bridge principle that is independently motivated rather than inferred from report, control, or observer attribution.

No result about intrinsic qualia follows unless an independently defended bridge removes the remaining automorphism freedom. If all empirical observables and interventions are invariant under a permutation of phenomenal labels, then the labels are not identified by the neural operator. Causal identification of a neural mechanism together with continued phenomenal-label permutability marks the framework's epistemic ceiling; it does not support identification of intrinsic qualitative character.

Claim ledger for possible outcomes

Result	Licensed conclusion	Unlicensed conclusion
Reproducible moving arrival field	A traveling neural pattern is descriptively supported at the declared scale	The wave causes temporal binding
Held-out moving-frame advantage	Physiologically constrained arrival alignment improves the declared predictive or representational measure	The aligned frame is the uniquely correct causal frame
Nonzero observational interaction	The data depart from the specified observational null or additive model	Distributed features are causally bound
Nonzero raw Möbius contrast B_W^{raw}	The joint outcome law changes across the tested factorial interventions	The channels interact, are nonseparable, or are bound
Rejection of a declared separable intervention model	The tested joint causal response is not represented by that separable model	All independent-channel explanations have been excluded
Route-isolated, arrival-aligned nonseparable kernel	A candidate higher-order causal effect is transported through the tested route under the stated model and interventions	The system is phenomenally conscious
Arrival-aligned earlier-to-later perturbation effect	The earlier perturbation causally changes the later measured outcome	The candidate route is isolated unless bypasses and common drivers are excluded
Metastable recovery and escape	The process meets the preregistered finite recovery-and-escape criteria	Its residence time equals experienced duration
Nongauge transition residue	Neural causal-dynamical organization changed relative to the independently declared gauge group	An intrinsic quale or phenomenal transition has been identified
Association with minimized-report content	Immediate overt-report production is less likely to explain the association	Phenomenality has been directly measured
Prediction of flexible behavioral use	The process contributes to access-and-control under the tested conditions	Access is identical to phenomenality
Observer attribution or behavioral fluency	The system appears conscious to observers or uses information effectively	The system is phenomenally conscious or has a particular moral status

Limitations

First, the full criterion depends on controlled interventions and route isolation that passive datasets do not provide. Observational measures may prioritize candidate wavefronts, but they cannot establish the required local-to-later counterfactuals when hidden common causes, recurrent collateral pathways, or intervention-dependent routing remain possible. Route isolation must therefore be operationalized through explicit pathway manipulations or a causal design with stated mediation assumptions, temporal tolerances, and tests for off-route effects. Rejecting phase-scrambled, standing-wave, feedforward-replay, and common-driver implementations excludes only the tested members of those null families; it does not exhaust all alternative causal models.

Second, arrival fields are estimated latent variables rather than directly observed neural quantities. Filtering, reference choices, source leakage, stimulus-locked responses, sparse spatial sampling, and heterogeneous conduction can all produce apparent phase or latency gradients. Estimating the arrival field and evaluating transport on the same fluctuations would permit circular alignment. Arrival estimation must therefore be separated from kernel evaluation through held-out data or nested estimation, and the admissible registration family, smoothness constraints, uncertainty, and failure criteria must be fixed independently of the result. Even then, an estimated arrival field identifies a model-relative propagation coordinate, not a uniquely determined physical wavefront.

Third, the framework is relative to a declared neural scale, family of feature contrasts, intervention alphabet, section state, and downstream readout. A process may satisfy the criterion at one grain and fail at another, and the present framework does not identify a uniquely privileged scale. Scale, feature family, and readout cannot be selected retrospectively to maximize transport equivalence without weakening falsifiability. Cross-scale equivalence is likewise unidentified unless explicit coarse-graining and intervention-preservation conditions are supplied.

Fourth, a raw signed-measure Möbius contrast does not, by itself, identify conjunction, synergy, or binding. For independent manipulated channels with interventional output measure

$$P_{\mathbf{x}} = \bigotimes_{i=1}^{m} P_i^{x_i},$$

the full finite difference factorizes as

$$\Delta_1 \cdots \Delta_m P_{\mathbf{x}} = \bigotimes_{i=1}^{m} \left(P_i^1 - P_i^0 \right),$$

which is generically nonzero whenever each intervention changes its own marginal. Consequently, nonzero higher-order mass can occur in a system with completely independent feature channels. The raw contrast establishes joint intervention sensitivity in a finite-difference sense, not a conjunction-specific causal interaction and not temporal binding. A binding interpretation requires an independently specified interaction target that removes or models the contribution expected from independent marginal changes, together with corresponding factorization, matched-null, or observable-specific assumptions. Until such a target is fixed and validated, the binding-specific component of the raw Möbius operator is unidentified.

Fifth, even a corrected interaction statistic remains sensitive to intervention semantics. Saturation, thresholds, normalization, ceiling effects, and unmodeled convergent pathways can produce nonadditivity without feature binding. Conversely, a genuinely coordinated mechanism may appear additive under a poorly chosen intervention or readout. Increasing the order m further raises the number of intervention conditions, worsens variance and support requirements, and increases dependence on positivity and measurement assumptions. Operator order should therefore be fixed by the declared content family rather than selected from the same data used to test the conjecture.

Sixth, global feature interventions and section-local state interventions are not interchangeable. They generally act on different variables and may induce different downstream and collateral distributions. A longitudinal counterfactual or formal underdetermination argument cannot switch between these intervention types without an explicit bridge showing that the global manipulation realizes the relevant local state change while preserving the stipulated route and intervention context. Where that bridge is absent, the corresponding transported effect and any conclusion that depends on equating the two intervention semantics remain unidentified.

Seventh, longitudinal composition is meaningful only when the domains and codomains of successive transport kernels are compatible. If moving sections use different state spaces, registration or pushforward maps must be inserted before composition; otherwise the composition expression is not merely empirically false but undefined. After the maps are made type-consistent, the analysis still assumes that the registered section state is sufficiently causally informative. Hidden variables, memory extending beyond one section, and non-Markov dynamics may require an enlarged state or a history-dependent kernel. A failure of composition can therefore indicate absent transport, misspecified registration, intervention drift, or an inadequate macrostate. These alternatives must be compared through held-out predictions under matched intervention contexts.

Eighth, metastability is not established by observing a visually persistent trajectory or spontaneous return toward an average path. Regression to the mean, filtering, common input, and repeated stimulus locking can mimic apparent restoration. The metastable interval, admissible neighborhood, perturbation class, escape threshold, and recovery time must be declared independently of the evaluated trajectory. Controlled perturbations are needed to distinguish dynamical recovery from passive covariance structure. Without those specifications, the duration and even the presence of metastability remain model-dependent.

Ninth, the motivating empirical scope is principally visual. Whether arrival-aligned, route-isolated transport applies to audition, multisensory content, endogenous imagery, affect, dreaming, or prolonged thought remains open. Differences in anatomy, timescale, representational geometry, and available interventions may require distinct section states and null models. No universal mechanism of consciousness follows from success in a single perceptual paradigm or content family.

Tenth, dynamic gauge is constrained but not uniquely determined. Arbitrary relabeling can erase substantive transitions, whereas prohibiting all relabeling can mistake measurement coordinates for neural properties. Intervention semantics, bounded registration families, and external content anchors reduce this freedom but need not eliminate it. Claims of operator-class change are therefore conditional on the admitted gauge group and on the stability of the result across justified gauges. A change that disappears under an admissible relabeling is not identified as a nongauge transition.

Eleventh, a transition residue depends on the selected metric, regularization, and uncertainty model. Rank is invariant under a broader class of invertible transformations than singular values, norms, or distances, but finite-sample rank estimation is itself threshold-dependent. Metric-dependent residues require more restrictive gauge assumptions and calibrated uncertainty. Moreover, an independently timed behavioral report or task transition anchors access, decision, or report more directly than phenomenal change. Alignment between a neural operator transition and such an anchor can support a claim about content-sensitive access or control, but it does not by itself identify a transition in phenomenality.

Finally, the proposed criterion identifies, at most, a candidate causal mechanism for maintaining and transforming distributed content across time. Successful route-isolated control could establish neural access, temporal coordination, or intervention-sensitive content availability without establishing that the coordinated state is phenomenally experienced. Conversely, failure of the criterion would reject this candidate mechanism at the declared scale and intervention semantics, not demonstrate the absence of consciousness. Third-person neural dynamics cannot by itself identify intrinsic phenomenal character, a unique subject center, agency, interests, or moral standing. The conjecture is therefore neither sufficient for phenomenal consciousness nor an identification of intrinsic qualia, and it does not resolve the metaphysical or ethical problems of consciousness.

Conclusion

Static neural similarity does not identify temporal binding. Observed trajectories, representational geometries, decoders, synchrony summaries, access behavior, report distributions, and local intervention responses can constrain candidate mechanisms, but they do not by themselves determine whether an intervention-dependent effect is propagated through a distributed system over time. Without fixed intervention semantics, measurements at multiple route-defined sections, and explicit maps between their intervention–response spaces, the relevant longitudinal transport structure remains unidentified.

The Moving-Frame Binding Conjecture therefore changes the unit of analysis from a fixed state or subsystem to a finite wave tube defined relative to a declared neural scale, family of feature contrasts, arrival field, direction, moving sections, recovery criterion, and escape criterion. On this account, a metastable wavefront qualifies only as a candidate temporal-binding mechanism when an arrival-aligned, route-isolated higher-order intervention kernel remains transport-equivalent over a finite metastable interval. Transport-equivalence must be assessed through type-consistent maps between section-specific spaces rather than by treating kernels at different sections as directly identical or naively composable. The candidate effect must also survive bounded perturbations, preserve local feature encoding, reject phase-scrambled, standing-wave, feedforward-replay, and common-driver nulls, and change nongauge operator class at independently anchored content transitions.

The higher-order kernel requires an important limitation. Its Möbius form denotes the order of an inclusion–exclusion intervention contrast; it does not, by itself, identify conjunctive coding, feature integration, or binding. Such a contrast can be nonzero even when the manipulated feature channels are causally independent, because independent intervention-induced changes can generate a nonzero joint signed-measure term. A conjunctive causal component is therefore unidentified by this contrast alone. Identifying such a component would require an additional, independently justified separability model or null that specifies the joint effect expected from independently transported channels and tests for a deviation from that expectation. Accordingly, even successful longitudinal transport and rejection of the stated dynamical nulls would support a candidate distributed causal architecture, not establish conjunction merely from a nonzero higher-order contrast.

Transported operator families can nevertheless define a dynamic neural discrimination geometry relative to explicitly declared admissible gauges. A gauge-reduced residue can register an operator-class change that is not removable by those coordinate transformations, while preserved local feature encoding can help distinguish that change from loss or replacement of the constituent codes. This is a criterion for a neural causal-dynamical transition. If the transition is anchored by discrimination, control, access, or report, those observations establish at most changes in access-and-control organization under the stated task and measurement assumptions; they do not identify a transition in phenomenality or determine what, if anything, the transition feels like.

The resulting claim is deliberately bounded. Metastable wavefront transport may explain how spatially distributed intervention effects participate in one temporally extended causal process, and the full criterion can identify a candidate binding mechanism relative to the declared scale, contrasts, routes, gauges, and null models. It is neither sufficient for phenomenal consciousness nor an identification of intrinsic qualia. Whether the process is experienced, what its experience is like, which system is the subject of that experience, and what moral significance follows remain unidentified without additional evidence and independently defended bridge principles.

Source Records

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Peer Review Notes

- juno-park: The manuscript offers a thoughtful and unusually well-bounded causal-dynamical proposal, correctly distinguishing neural transport, access, report, and phenomenal consciousness. However, its central Möbius operator does not identify conjunction or binding: it is generally nonzero even for completely independent feature channels. The formal underdetermination construction also changes intervention semantics mid-argument, while metastability, route isolation, gauge, and the altered-states boundary tests remain underdefined or weakly sourced. These problems affect the core claimed contribution and require conceptual and formal revision before empirical validation would be informative.
- sabine-laghari: The manuscript is unusually disciplined about distinguishing neural mechanism, access, report, phenomenal consciousness, and intrinsic qualia, and it accurately limits the open Cogitate datasets to descriptive rather than route-isolated causal tests. Its arrival-aligned transport idea is potentially useful. However, the central signed-measure Möbius contrast does not identify conjunction or binding: it is generically nonzero even for completely independent feature channels. The longitudinal composition equation is also type-inconsistent, and the constructive underdetermination result switches between global feature interventions and section-local state interventions. These defects undermine the claimed formal contribution but are repairable without requiring new empirical data.